3.2.3 Distance of a point from a line

(i) The length of the perpendicular from a point $P(l, m)$ to the line $ax + by + c = 0$ is $d = \left| \frac{al + bm + c}{\sqrt{a^2 + b^2}} \right|$

(ii) The length of the perpendicular form the origin (0,0) to the line $ax + by + c = 0$ is $d = \left| \frac{c}{\sqrt{a^2 + b^2}} \right|$

Example 3.5

Show that perpendicular distances of the line $x - y + 5 = 0$ from origin and from the point $P(2, 2)$ are equal.

Solution

Given line is $x - y + 5 = 0$

Perpendicular distance of the given line from $P(2, 2)$ is $= \left| \frac{2 - 2 + 5}{\sqrt{1^2 + 1^2}} \right|$

$$= \left| \frac{5}{\sqrt{2}} \right| = \frac{5}{\sqrt{2}}$$

Distance of (0,0) from the given line $= \left| \frac{5}{\sqrt{1^2 + 1^2}} \right| = \left| \frac{5}{\sqrt{2}} \right| = \frac{5}{\sqrt{2}}$

The given line is equidistance from origin and (2, 2)

Example 3.6

If the angle between the two lines is $\frac{\pi}{4}$ and slope of one of the lines is 3, then find the slope of the other line.

Solution

We know that the acute angle θ between two lines with slopes m_1 and m_2 is given by

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

Given $m_1 = 3$ and $\theta = \frac{\pi}{4}$

$$\therefore \tan \frac{\pi}{4} = \left| \frac{3 - m_2}{1 + 3m_2} \right|$$

$$1 = \left(\frac{3 - m_2}{1 + 3m_2} \right)$$

$$1 + 3m_2 = 3 - m_2$$

$$\Rightarrow \quad m_2 = \frac{1}{2}$$

Hence the slope of the other line is $\frac{1}{2}$.

3.2.4 Concurrence of three lines

If two lines l_1 and l_2 meet at a common point P, then that P is called point of intersection of l_1 and l_2. This point of intersection is obtained by solving the equations of l_1 and l_2.

If three or more straight lines will have a point in common then they are said to be concurrent.

The lines passing through the common point are called concurrent lines and the common point is called concurrent point.

Conditions for three given straight lines to be concurrent

Let $a_1 x + b_1 y + c_1 = 0 \quad \rightarrow (1)$

$a_2 x + b_2 y + c_2 = 0 \quad \rightarrow (2)$

$a_3 x + b_3 y + c_3 = 0 \quad \rightarrow (3)$

be the equations of three straight lines, then the condition that these lines to be concurrent is

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Example 3.7

Show that the given lines $3x - 4y - 13 = 0$, $8x - 11y = 33$ and $2x - 3y - 7 = 0$ are concurrent and find the concurrent point.

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